

CIND 820: Big Data Analytics Project

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Date of Submission: 2026-05-18



Abstract

Title: Insurance Fraud Detection Using Machine Learning on Claims Data

With the rapid advancement of digital technologies and data-driven decision-making, the insurance industry is increasingly adopting analytics and artificial intelligence to improve operational efficiency and reduce financial losses. One of the major challenges faced by insurance companies today is fraudulent insurance claims, which cost the industry billions of dollars annually and negatively impact both insurers and policyholders. As the volume and complexity of insurance claims continue to grow, traditional rule-based fraud detection methods are becoming less effective in identifying suspicious activities in real time.

This project focuses on developing a machine learning-based fraud detection system using publicly available insurance claims data. By analyzing patterns, anomalies, and behavioral characteristics within claim records, the study aims to identify factors associated with fraudulent claims and evaluate the effectiveness of predictive models in detecting potential fraud.

Various data analytics and machine learning techniques such as classification modeling, anomaly detection, and exploratory data analysis will be utilized to improve fraud identification accuracy and support data-driven decision-making within the insurance sector.

The findings of this project may help insurance organizations strengthen fraud prevention strategies, optimize claim investigation processes, and minimize financial risks while improving operational efficiency. Furthermore, the project demonstrates the growing importance of data analytics and machine learning applications in modern insurance systems and risk management practices.

Professional Context and Problem Definition

This capstone project aligns with the role of a Data Analyst / Fraud Analytics Specialist within the insurance industry. The project simulates a real-world business environment where analytics professionals use historical insurance claims data, statistical analysis, and machine learning techniques to identify potentially fraudulent claims and support risk management operations.

Problem Context:

Insurance fraud is a major challenge for insurance providers, resulting in billions of dollars in financial losses each year. The Coalition's study shows that \$308.6B is stolen every year in Insurance fraud in the United States. [\[1\]](#)

Fraudulent claims increase operational costs, delay legitimate claim processing, and negatively impact customer trust and premium pricing. Traditional rule-based fraud detection systems often struggle to identify evolving fraud patterns and complex fraudulent activities efficiently.

The purpose of this project is to develop a predictive analytics solution that can analyze insurance claims data and identify suspicious claims that may indicate fraudulent behavior. The project will focus on understanding the patterns, characteristics, and risk indicators associated with fraud while evaluating the effectiveness of machine learning models for fraud detection and decision support.

Key Beneficiary:

The primary beneficiaries of this project are:

- Insurance companies and fraud investigation departments
- Risk management and underwriting teams
- Claims processing analysts
- Business intelligence and analytics teams
- Policyholders who may benefit from reduced fraud-related costs and fairer premium

The project may also provide value to organizations seeking to improve operational efficiency and strengthen data-driven fraud prevention strategies.

Real World Significance of the Project:

Insurance fraud remains a significant financial and operational issue across the global insurance industry. Detecting fraudulent claims accurately and efficiently can help insurance companies reduce unnecessary losses, improve claim processing workflows, and allocate investigative resources more effectively. As fraud schemes become increasingly sophisticated, organizations are adopting data analytics and machine learning technologies to strengthen fraud detection capabilities.

This project demonstrates the practical application of data analytics, predictive modeling, and anomaly detection techniques within a real-world insurance context. The insights generated from the analysis may help support faster decision-making, improve fraud prevention strategies, and contribute to more efficient and cost-effective insurance operations.

Additionally, the project highlights the growing importance of data analytics in modern risk management and business decision-making processes.

Research Questions:

1. What patterns and characteristics are most commonly associated with fraudulent insurance claims?
2. How accurately can machine learning models predict potentially fraudulent insurance claims using historical claims data?
3. Which variables contribute most significantly to fraud prediction and decision-making?
4. How can data-driven fraud detection improve operational efficiency within insurance claim processing?

These research questions are important because insurance fraud represents a major financial and operational challenge for insurance providers worldwide. Fraudulent claims contribute to billions of dollars in annual losses, increased insurance premiums, and inefficiencies in claims management processes. Traditional fraud detection methods are often limited in their ability to identify evolving fraud patterns and large-scale suspicious activities.

By exploring the relationship between claims data and fraudulent behavior, this project aims to demonstrate how data analytics and machine learning can support faster, more accurate, and scalable fraud detection systems. The research questions are interconnected because they collectively address fraud identification, predictive modeling performance, operational decision-making, and business value generation within the insurance industry.

Impact Hypothesis:

It is hypothesized that applying machine learning and predictive analytics techniques to insurance claims data will significantly improve the ability to detect potentially fraudulent claims compared to traditional rule-based approaches. By identifying high-risk claims earlier in the claims process, insurance companies may reduce financial losses, improve operational efficiency, and allocate investigative resources more effectively.

The outcomes of this project may contribute to stronger fraud prevention strategies, enhanced risk management practices, and more data-driven decision-making within the insurance industry. From a societal perspective, improved fraud detection can help reduce unnecessary costs passed on to policyholders through increased insurance premiums while promoting fairness and transparency in insurance operations.

Data Source:

This project will use the publicly available Vehicle Insurance Claim Fraud Detection Dataset obtained from Kaggle. The dataset contains historical automobile insurance claim records designed for fraud detection and predictive analytics research.

The dataset includes information related to:

- Policyholder demographics
- Vehicle characteristics
- Incident and accident details
- Claim amounts
- Policy information
- Fraud labels indicating whether a claim is fraudulent or legitimate

The dataset is suitable for classification and anomaly detection tasks and supports the development of machine learning models for insurance fraud prediction.

Dataset:

The Project will utilize the below dataset for analysis and for machine learning model development and training. [2]

<https://www.kaggle.com/datasets/shivamb/vehicle-claim-fraud-detection/data>

Privacy and Ethics Audit:

The dataset does not contain direct personally identifiable information such as:

- Names
- Addresses
- Phone numbers
- Email addresses
- Social Insurance Numbers
- Banking or payment details

The data appears to be anonymized and structured for research purposes. This reduces privacy risks and makes the dataset appropriate for academic research and analysis.

The dataset is publicly available on Kaggle for educational and research purposes. The project will comply with Kaggle's dataset usage policies and academic integrity requirements.

The dataset will only be used for:

- Educational analysis
- Academic research
- Non-commercial machine learning experimentation

Proper attribution to the dataset source and platform will be included throughout the capstone report and presentation.

Limitations of the Research:

The scope of this study is limited to a publicly available insurance claims dataset and the historical records it contains. As a result, the findings may not fully represent the complexity, diversity, or evolving nature of real-world insurance fraud patterns across different insurers, regions, or time periods. The machine learning models used in this analysis are trained on historical claim data, and therefore their predictive performance may be affected by changes in fraud tactics, policy structures, and operational processes over time.

Additionally, the accuracy and reliability of the results depend heavily on the quality, completeness, and representativeness of the dataset. Since the dataset may contain simplified, anonymized, or pre-processed variables, certain real-world contextual factors used by insurance companies in fraud investigations may not be precisely captured. External influences such as regulatory changes, economic conditions, and emerging fraud strategies are not included in the dataset, which may limit the generalizability of the model outcomes in real operational environments.

Proposed Technical Pipeline:

1. Data Ingestion:

The project begins with importing the insurance claims dataset from Kaggle into a structured analytical environment (Python) where Initial data profiling will be performed to understand schema, data types, missing values, and distribution of key variables such as claim amount, incident details, and fraud labels.

2. Data Cleaning & Preprocessing:

This stage involves handling missing values, encoding categorical variables, detecting outliers, and normalizing numerical features where necessary. Feature engineering may also be applied to derive meaningful variables that improve fraud detection performance.

3. Exploratory Data Analysis (EDA):

EDA will be conducted to identify trends, correlations, and patterns associated with fraudulent and non-fraudulent claims. Visualization techniques such as heatmaps, histograms, and boxplots will be used to uncover insights and guide feature selection.

4. Machine Learning Model Development:

Multiple classification algorithms such as Logistic Regression, Decision Trees, Random Forest, and XGBoost will be implemented to predict fraudulent claims. Models will be trained on historical data and optimized using hyperparameter tuning techniques.

5. Model Evaluation & Validation:

Model performance will be assessed using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Special emphasis will be placed on recall to minimize missed fraud cases.

6. Insights & Visualization:

Key findings will be translated into business insights using dashboards and visualizations (Power BI). Feature importance analysis will help explain model decisions and highlight fraud risk drivers.

All stages are interconnected in an end-to-end workflow where insights from EDA inform feature engineering, processed data feeds machine learning models, and model outputs are translated into actionable business intelligence.

The proposed pipeline relies on a single structured dataset, widely available Python libraries, and standard machine learning workflows. Each stage is modular, allowing incremental progress and parallel work on analysis, modeling, and reporting to ensure timely completion of the capstone project.

GenAI Declaration:

Artificial Intelligence tools, such as ChatGPT, may be used in this capstone project strictly as a supportive tool to enhance learning and productivity. AI tools will be used for purposes such as improving clarity of written content, assisting with structuring the report, generating alternative wording for technical explanations, and supporting brainstorming of analytical approaches.

However, all analytical work, including data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and interpretation of results, will be independently conducted. AI tools will not be used to generate final analytical results, conclusions, or model outputs without validation.

All content generated with AI assistance will be carefully reviewed, edited, and adapted to ensure accuracy, relevance, and alignment with academic requirements.

References

- [1] Coalition Against Insurance Fraud, "The Impact of Insurance Fraud," Coalition Against Insurance Fraud, Washington, DC, USA. [Online]. Available: [Coalition Against Insurance Fraud](#). [Accessed: May 16, 2026].
- [2] Kaggle, "Vehicle Claim Fraud Detection Dataset," Kaggle Inc. [Online]. Available: [Kaggle Vehicle Claim Fraud Dataset](#). [Accessed: May 17, 2026].